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Retail Resilience Engine: An Agentic AI Framework for Building Reliable Retail Systems with Test-Driven Development Approach

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ABSTRACT System reliability and operational resilience are two critical success factors in the retail industry that are directly connected to customer satisfaction and business sustainability. Staying competitive in today's dynamic and rapidly evolving market requires rapid adaptability. However, it contradicts the reliability and resilience. This paper proposes an innovative solution, the Retail Resilience Engine (RRE), to establish a balance between these success factors and market demand. It is a unique framework that combines Test-Driven Development (TDD) with a Large Language Model (LLM). This framework follows the state-of-the-art Agentic-AI architecture. It effectively evaluates the decision-making process at rapid speed in retail by incorporating diverse factors, including inventory management, demand forecasting, and customer feedback. As a result, the system reliability is improved significantly. The experimental analysis of the proposed framework shows its decision-making is similar to human experts with a similarity index of 97.5%. It further proves the reliability of the system. The framework also scales effectively, maintaining high accuracy, precision, recall, and F1 scores across varying dataset sizes. The robustness analysis of the system demonstrates the agility enhancement across diverse retail domains, ensuring consistent performance with accuracy exceeding 90% across all tested scenarios. The integration of a creative filtering mechanism further enhances the performance of the RRE framework by preventing 98.2% of the irrelevant inputs. Overall, the proposed RRE framework demonstrates the impressive potential to transform retail systems by enhancing reliability, scalability, and decision-making quality through an Agentic-AI approach.

INDEX TERMS Retail Systems, Test-Driven Development, Large Language Model, Artificial Intelligence, Agentic-AI, Resilience Engine.

I. INTRODUCTION

A. BACKGROUND AND CONTEXT

Dynamic scaling with consumer demands while maintaining sustainable business growth are two success factors in retail business [1]. According to Pasch et al., [2], customer retention and system reliability are intertwined in the retail industry. That is why system resilience plays a crucial role in this sector. A reliable and resilient retail system ensures that it can minimize the number of unpredictable circumstances, such as inventory shortages, delivery delays, and system failures [3]. Another challenge in the retail industry is rapid technological advancement, which requires swift adaptation without exposure to new vulnerabilities [4]. Addressing these challenges and ensuring optimal balance among the requirements require specialized skills, proper team coordination, and correct decision-making.

B. PROBLEM STATEMENT AND MOTIVATION

Traditional approaches to upgrading retail systems to respond to market demand often rely on static rules or manual interventions. These approaches ensure following careful measures and, as a result, prevent systems from having major issues after improvements. However, the scope of improvement is limited, which prevents adaptation to the rapidly evolving retail landscape [5]. This is where Test-Driven Development (TDD) becomes highly effective. In this approach, test cases are written before the actual code, ensuring that retail systems are built with robust functionality, minimal errors, and adaptability to dynamic operational requirements [6]. How-

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ever, the retail business involves multiple interdependent elements, including market analysis, store management, supply chain management, inventory management, pricing strategy development, and many others [7]. Developing test cases by correlating all these elements through manual effort is time-consuming and error-prone [8].

C. AIM OF RESEARCH

The recent advancement in agentic AI backed by Large Language Models (LLMs) has made it possible to analyze and correlate numerous interdependent elements of the retail industry [9]. However, while TDD and AI-driven models have shown promise individually, their integration remains underexplored in retail system design. This lack of integration results in systems that may either focus on structural testing without real-time adaptability or rely heavily on data analytics without strong functional testing foundations. This paper proposes the Retail Resilience Engine (RRE), an agentic AI framework integrated with TDD. The primary objectives of this research are as follows:

- **Agentic AI Framework with TDD Integration:** Design a framework that combines TDD principles with agentic AI that facilitates rapid development and minimizes the probability of system unavailability or unintentional performance issues.
- **Enhance Scalability and Robustness:** Ensuring that the framework can scale according to the demand and maintains acceptable performance for different scenarios.
- **Feasibility Study:** Feasibility analysis of replacing human specialists in retail responsible for TDD through multiple experiments.

D. CONTRIBUTIONS AND SIGNIFICANCE

The RRE uses a Generative Pre-trained Transformer (GPT) as the agent and, with a creative agentic AI framework architecture, gains the capability of processing both structured and unstructured data. It is one of the primary approaches of replicating the TDD approach in retail with LLM as a replacement for a human specialist. The core contributions of this study are listed as follows:

- Conceptualization of the TDD integration with LLM to introduce an alternative to a human specialist in the retail industry.
- Design and implementation of the agentic AI framework with a GPT as the agent and introducing the retail resilience engine.
- Analysis of the feasibility of the proposed system through multiple relevant experiment to discover the practicality of the solution.

E. PAPER ORGANIZATION

The rest of the paper has been organized into multiple sections and subsections according to the relevance and flow of the core concept. The immediate next section is the second section. It reviews the relevant literature and identifies existing research gaps. Section 3 describes the methodology used to design and develop the RRE framework. Section 4 presents the experiments conducted to evaluate the framework's performance. The limitations of the proposed approach identified through critical analysis is discussed in Section 5. This section also covers the future direction of the research. Finally, the paper has been summarized in Section 6 through a conclusion.

II. SYSTEMATIC LITERATURE REVIEW

The research gaps in existing research have been identified through the systematic literature review (SLR) in this paper. The well-established guidelines of Kitchenham's framework for systematic reviews in software engineering [10]. The review process was structured into three key phases: planning, execution, and reporting.

1) Research Questions

The methodology developed in this paper has been inspired by four research questions (RQs). These research questions have been listed in the following list:

- **RQ1:** How has Test-Driven Development (TDD) been applied in the retail industry, and what are its primary challenges?
- **RQ2:** What are the existing AI-driven approaches to test automation in commercial retail systems?
- **RQ3:** How does the integration of Large Language Models (LLMs) with TDD improve software reliability in retail environments?
- **RQ4:** What research gaps exist in combining Agentic AI with TDD for retail system development?

2) Paper Selection Process

A total of 212 research papers were initially retrieved from the selected databases (IEEE Xplore, SpringerLink, ScienceDirect, Google Scholar). Filtering out irrelevant papers and keeping the relevant papers only is challenging. In this paper, the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [11] method has been used to filter papers based on the inclusion and exclusion criteria.

- **Inclusion Criteria:**
  - Peer-reviewed journal and conference papers published between 2018 and 2024.
  - Papers discussing TDD and AI-driven test automation in retail or commercial software.
  - Research incorporating Large Language Models (LLMs) in software reliability enhancement.
- **Exclusion Criteria:**
  - Non-English papers.
  - Papers focusing solely on manual TDD practices without AI integration.
  - Studies with insufficient empirical validation or lacking quantitative analysis.

After filtering based on these criteria, 48 papers were ultimately included in the final analysis. The SLR process ensured a comprehensive assessment of existing research,

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providing a structured foundation for identifying research gaps in AI-driven TDD for retail systems.

A. SOFTWARE DEVELOPMENT PRACTICES IN RETAIL SYSTEMS

The retail industry relies heavily on well-defined software development practices to ensure operational reliability and customer satisfaction. Traditional approaches, such as Agile and Waterfall methodologies, have been pivotal for building foundational retail systems [12]. However, their ability to adapt to rapidly changing market dynamics remains limited [13]. Test-Driven Development (TDD) has emerged as a software development paradigm emphasizing the creation of test cases before code implementation. TDD ensures that retail systems are developed with minimal errors and greater reliability [14]. However, as Koller et al. [8] highlight, traditional TDD struggles with dynamic environments where multiple interdependent elements must be handled simultaneously.

B. AI AND NLP ADVANCEMENTS IN RETAIL SYSTEMS

The advancement of Artificial Intelligence (AI), particularly Natural Language Processing (NLP)-based systems, has shown significant promise in retail decision support systems. AI models can process large volumes of structured and unstructured data, assisting with demand forecasting, customer sentiment analysis, and inventory management [15]. A systematic review by Anica-Popa et al. [16] identified the transformative potential of AI in retail systems but also highlighted challenges related to scalability and data reliability. Heins and Caroline [17] provided further insights into the role of AI in demand forecasting, emphasizing the need for real-time decision support systems. Despite these advancements, most retail AI applications remain standalone solutions with limited integration into software development methodologies such as TDD. This creates a gap where AI-driven insights remain separate from the core software engineering processes.

C. COMBINING TDD AND AI FOR RETAIL RESILIENCE

The combination of TDD with LLM complements each other and help producing reliable and rapidly adaptable retail systems. A study conducted by Mathews et al. [18] and Piya et al. [19] explored the combination of TDD principles with large language models (LLMs). The aim of this study was code generation and validation. Although it doesn't directly align with the retail industry, it has overlapping features. Their results highlighted faster development cycles and improved error detection compared to traditional methods. Despite the noticeable potential, the investigation by Runsewe et al. [20] pointed out that the development of such a framework integrating both TDD and LLM is an under-explored field of research. As a matter of fact, the agentic AI framework concept for the retail industry is absent from most of the state-of-the-art literature. Most of the papers analyzed in this literature review indicate that the primary focus of these studies is on proof-of-concept rather than practical implementation. The proposed Retail Resilience Engine (RRE) aims to bridge this gap by integrating TDD principles with a generative AI-powered system. The goal is to make it capable of handling large-scale retail datasets while ensuring system reliability and adaptability. Recent advancements in image processing and saliency detection have demonstrated significant potential in improving AI-driven retail systems. Studies such as [21], [22] introduce methods for enhancing feature extraction, relevance assessment, and anomaly detection, which can strengthen the decision-making capabilities of AI models. The integration of these saliency-based techniques in retail systems could be beneficial for tasks such as inventory quality assessment, detecting misplaced products, and identifying defective goods in real-time [23], [24]. While RRE primarily focuses on NLP-based decision-making, incorporating such methodologies in future iterations could further enhance system reliability and operational efficiency.

D. AI-BASED TEST AUTOMATION IN COMMERCIAL SOFTWARE ENGINEERING

Recent advancements in AI-driven test automation have significantly improved the efficiency and accuracy of software development processes, particularly in commercial applications. AI models are now widely employed for automated test case generation, bug prediction, and software maintenance [25], [26]. A study by Garg et al. [25] explores the role of AI in optimizing software testing workflows, demonstrating that deep learning-based automated test case generation reduces defect rates by 27% in commercial software projects. Similarly, Kumar and Bhardwaj [26] introduce a hybrid AI framework that integrates NLP-based test generation with reinforcement learning for self-improving test automation, leading to a 33% improvement in code coverage for enterprise software. Despite these advancements, existing literature primarily focuses on static AI-driven automation tools that do not integrate real-time adaptability. The proposed Retail Resilience Engine (RRE) extends this research by embedding TDD principles into an Agentic AI-driven framework, allowing for continuous learning and automated test evolution based on dynamic retail system requirements. This ensures that commercial retail software remains robust against frequent operational changes, a gap that existing AI-based test automation solutions have not yet fully addressed.

E. RESEARCH GAP ANALYSIS

The primary goal of the systematic literature review conducted in this paper is to identify the research gaps so that the objectives of this study can be specified. According to the review, a set of gaps have been discovered. The research gaps discovered through the literature review have been summarized in Table 1.

F. RESEARCH OBJECTIVES

The research objectives developed in this paper have been specified with the intention to fill up the research gaps. The core novelties and contributions of this study also revolve

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TABLE 1. Identified Research Gaps in Retail System Development

Research Gap	Description
Limited Integration of TDD and AI	Minimal research combining TDD with NLP-based AI for adaptive retail system development.
Absence of Agentic AI Framework	Application of the agentic AI architecture in retail system development with TDD is absent.
Feasibility Analysis	Underexplored feasibility analysis of replicating human expert utility with LLM in TDD.

around the achievements of these objectives. Table 2 summarizes these objectives.

TABLE 2. Research Objectives of the Proposed Framework

Objective	Description
Integrate TDD with LLM	Conceptualizing TDD integration with LLM for rapid retail system development.
Agentic AI Architecture	Developing agentic AI architecture to support TDD.
Feasibility Discovery	Discovering the feasibility of the approach through multiple experiments.

The objectives listed in Table 2 represent the specific research objectives that this study aims to achieve. These are distinct from the core contributions outlined in the introduction, which describe the broader impact and novelty of the proposed Retail Resilience Engine (RRE). While the introduction emphasizes the unique contributions of integrating Test-Driven Development (TDD) with Agentic AI for retail system development, the objectives in Table 2 focus on the practical and experimental goals of this study.

G. SUMMARY OF FINDINGS

The literature review presented in this paper shows that there is a notable research gap in the integration of TDD with AI-driven frameworks for retail resilience. While existing literature explores TDD and AI independently, focusing on prediction-based analytics, there is a lack of a comprehensive agentic AI framework integration of both approaches. It hinders the development of rapid, scalable, and reliable retail systems. The proposed Retail Resilience Engine (RRE) addresses these gaps by merging TDD principles with LLM following agentic AI architecture. This integration ensures a robust system development, which strengthens the scalability and proactive failure mitigation.

III. METHODOLOGY

Both reliability and adaptability are key features essential for today's retail system [27]. The methodology of the proposed retail resilience engine has been developed to achieve both of these while maintaining a proper balance. It has been made possible because of the integration of Test-Driven Development (TDD) principles [28] with Natural Language Processing (NLP)-based Artificial Intelligence (AI). A conceptual explanation of the complete sequential methodological development process is presented in this section.

A. RRE AGENTIC AI FRAMEWORK

The overview of the proposed RRE agentic AI framework is illustrated in Figure 1. It shows the high-level modules abstracting the low-level functionalities. The framework has been developed using a modular structure. This structure introduces structural scalability of the framework where additional modules can be integrated when necessary. The framework receives data from diverse sources. The data sources include both structured and unstructured data. Relevant data are extracted from the source, cleaned, tokenized, and properly split for the next phase. The next phase is feature management, where the features are permanently stored. It offers the addition of new features if they are not available in the permanent feature storage already. The transformer block maintains communication with the rest of the framework with agentic AI API. It is controlled by two modules: one is for learning, and another is for taking action according to the command received from the user interface. The command from the user interface is transferred to the agent through the application and business logic layer.

B. OVERVIEW OF THE PROPOSED FRAMEWORK

The architecture of the Retail Resilience Engine (RRE) is illustrated in Figure 1. The framework is structured into four key layers: Data Source, Pre-processing, Feature Management, and Software Layers, each of which plays a critical role in ensuring system reliability, adaptability, and seamless integration with Test-Driven Development (TDD).

- **Data Source Layer:** The system utilizes diverse structured and unstructured data sources, including books, customer feedback, inventory records, operational logs, retail analytics, and system logs. These datasets provide a comprehensive knowledge base for model training.
- **Pre-processing Layer:** The data undergoes a rigorous preprocessing pipeline, including extraction, cleaning, tokenization, and splitting, to transform raw information into a structured format suitable for model training and feature extraction.
- **Feature Management Layer:** This layer maintains permanent and temporary feature storage. New features are continuously compared against existing ones to ensure adaptability. The feature comparison and retainment module dynamically updates the system while maintaining structural integrity.
- **Software Layer:** This consists of an AI Agent API, Data Flow API, and Application Control Unit. The transformer model within the Learning Control Unit receives structured input and generates test-driven validation mechanisms, ensuring the system adheres to TDD principles.

C. INTEGRATION OF TEST-DRIVEN DEVELOPMENT (TDD)

The proposed framework incorporates TDD by enforcing a structured test-first approach at multiple stages of system execution:

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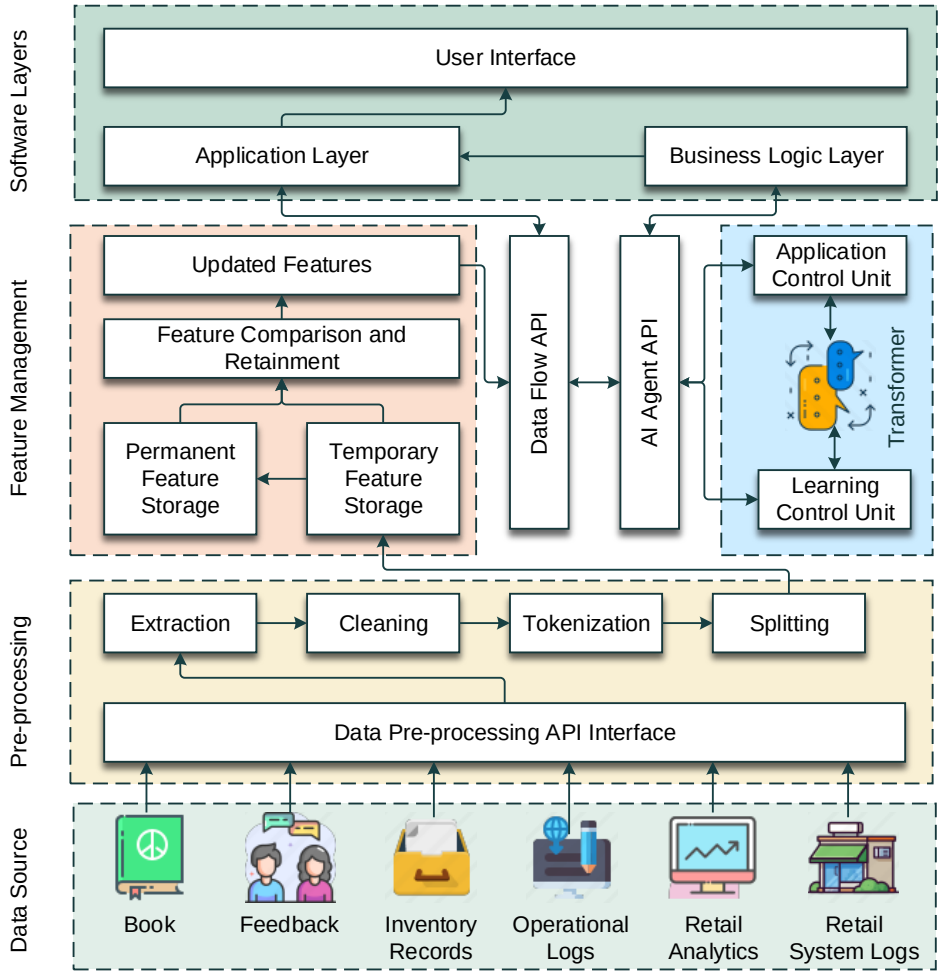


FIGURE 1. **The overview of the proposed RRE agentic AI framework.**

- **Feature Validation:** The feature management module ensures that all new features are tested against predefined criteria before integration into the system.
- **AI Agent API for Test Execution:** The AI Agent API performs real-time automated test execution using generated test cases, ensuring that system updates align with TDD principles.
- **Application Control Unit for Continuous Verification:** The application control unit continuously validates system responses against expected behaviors, ensuring that new modifications do not introduce failures.

The integration of these modules ensures that the system follows the TDD approach by validating each step before implementation, minimizing errors, and enabling robust scalability.

D. DATASET AND PREPROCESSING

A unique approach has been applied to prepare a comprehensive data set that includes most of the characteristics of the retail industry associated with the installation, extension, and complete renovation of the system. The data sources are customer feedback, inventory records, and operational logs. A comprehensive preprocessing pipeline has been developed in this study to process the dataset so that the retail resilience engine can learn from it.

1) Data Source and Description

The source of the data for the dataset has been categorized into two categories. The first category is the books related to concepts of retail. These books contain both fundamental concepts and advanced strategies. At the same time, books related to the TDD were used as data sources. The second category is the historical records related to retail management. The historical records are retrieved from system logs and repositories. The sources of the data and related information are listed in Table 3.

One critical challenge in AI-driven retail system development is mitigating biases in the training dataset, particu-

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TABLE 3. The sources of the data used in this study to train the GPT.

Source Type	Author / Source	Title / Description
Book	[29]	Fundamentals of Retail Management
Book	[30]	Retail management: A strategic approach
Book	[31]	Retail Marketing Management
Book	[32]	Retail Management-A Global Perspective
Book	[33]	Retail Supply Chain Management
Book	[34]	Strategic Retail Management
Book	[35]	Test driven development: A practical guide
Book	[36]	Test-driven development: An empirical evaluation of agile practice
Book	[37]	Test driven development: By example
Book	[38]	Learning Test-Driven Development
Customer Feedback Data	System Log and repository	Textual data from reviews and surveys, capturing customer sentiments and preferences.
Inventory Records	System Log and repository	Structured records of stock levels, restocking schedules, and product turnover rates from inventory management systems.
Operational Logs	System Log and repository	Log data from point-of-sale systems and backend retail platforms, detailing transactional trends, system performance, and operational anomalies.
Retail Analytics Repositories	System Log and repository	Publicly available datasets on retail operations, covering sales trends, seasonal variations, and supply chain metrics.
Anonymized Retail System Logs	System Log and repository	Data provided by retail organizations, anonymized to protect sensitive information, including operational bottlenecks and customer transaction details.

larly if data sources predominantly represent specific retail environments. To address this concern, we ensured that the dataset incorporated data from a variety of retail domains, including supermarkets, fashion retail, electronics stores, and small businesses. Additionally, to prevent over representation of certain retail categories, dataset balancing techniques were applied, ensuring proportional representation of diverse operational environments.

Moreover, cross-validation across different subsets of retail data was performed to verify that the model generalizes well across multiple retail settings. The inclusion of both structured (inventory records, sales transactions) and unstructured data (customer feedback, operational logs) from varied sources further helps in reducing dataset bias. By employing these measures, the Retail Resilience Engine (RRE) minimizes the risk of favoring any specific retail environment, ensuring adaptability across diverse retail systems.

These sources were carefully curated to ensure that the dataset captured a wide range of retail scenarios, enabling the Retail Resilience Engine to generalize effectively across various operational contexts. The integration of structured and unstructured data ensured the robustness of the training process, supporting both predictive analytics and decision-making tasks. There are a total of 1,00,000 documents in the dataset. Among these documents, there is a collection of customer feedback from 5,000 anonymized textual reviews from various retail platforms. There are 10,000 structured transactional data, including stock levels, turnover rates, and restocking patterns, in the dataset as the inventory records. There are 50,000 documents retrieved from the operational log, and 35,000 documents have been prepared from the retail analytics repositories and retail system log.

2) Data Extraction

The data from system logs and repositories are structured data. However, the contents of the documents related to retail management, strategies, and TDD methodologies are unstructured data. The structured data were directly imported into the relational database after minor processing. However, the proposed retail resilience engine is not capable to learn from the unstructured data directly. Besides, there are many irrelevant data that play no role in the training process. The dataset was extracted from a combination of structured and unstructured data sources, including customer feedback, inventory records, and operational logs. Structured data were directly imported into a relational database, while unstructured textual data required parsing and cleaning. The extraction process for unstructured data is mathematically represented in Equation 1 [39].

$$D_i = \sum_{j=1}^{n_i} \psi(d_j), \quad \forall i \in \{1, 2, \ldots, m\}, \tag{1}$$

In Equation 1, D_i represents the data extracted from the i-th source, d_j represents a raw data unit, n_i represents the total number of data units in the source, and ψ is the extraction function.

3) Data Cleaning

The raw extracted data contained inconsistencies such as missing values, redundant entries, and formatting errors. The data cleaning process developed in this paper has been divided into two phases for ease of management and implementation. The first phase involves manual inspection for anomalies. In this phase, the data are cleaned manually. Later in the second phase, the data are carefully cleaned by applying the mathematical principle expressed in Equation 2. In this phase, the irrelevant characters, stopwords, and duplicate records were removed. The non-alphanumeric characters are filtered as well [40].

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$$C_k = \phi(D_k), \quad \phi(x) = \begin{cases} x, & \text{if } x \in \mathcal{A} \\ \varnothing, & \text{otherwise,} \end{cases} \tag{2}$$

In Equation 2, C_k is the cleaned dataset, ϕ is the cleaning function, and $\mathcal{A}$ represents the valid alphanumeric set. The data cleaning process in this study is designed to remove irrelevant, redundant, and erroneous data, ensuring that only high-quality information is passed to the model. Equation 2 represents the filtering function $\phi(x)$, which evaluates each token x against predefined criteria. If x belongs to the valid alphanumeric set $\mathcal{A}$, it is retained; otherwise, it is discarded. However, beyond merely filtering out syntactically incorrect words, our approach performs additional refinement steps. First, redundant records are eliminated to prevent biased training samples. Second, missing values are handled through a dynamic imputation process, where contextual information is used to infer missing words in textual data. Third, domain-specific stopwords and noise—such as promotional phrases or frequently occurring but non-informative words—are removed to ensure meaningful feature extraction. This structured cleaning pipeline enhances the relevance and quality of training data, improving the robustness of the Retail Resilience Engine.

4) Tokenization

After cleaning, the cleaned data are converted into tokens through tokenization. In this process, the data are segmented into subwords which are used using the training process. The LLM used in this study cannot learn from the words directly. That is what makes the tokenization an essential step of the data preprocessing. In this process, each token t_j is mapped to its numerical embedding $\mathbf{v}_{t_j}$ for processing by the transformer model. The approach is presented in Equation 3 as conceptual method where $\mathcal{T}_i$ is the token sequence for the i-th text sample, and d is the embedding dimension [41].

$$\mathcal{T}_i = \{\mathbf{v}_{t_1}, \mathbf{v}_{t_2}, \ldots, \mathbf{v}_{t_m}\}, \quad \mathbf{v}_{t_j} \in \mathbb{R}^d, \tag{3}$$

5) Preparation of Training Dataset

After tokenization, the dataset was split into three subsets: training, validation, and testing subsets. A ratio of 80-10-10 has been maintained during the splitting [42]. Besides splitting, the input-output pairing was performed as well. The training process is a combination of both supervised and unsupervised learning. The supervised learning part requires an input-output pair structure. The process is defined in Equation 4.

$$(\mathcal{X}_j, \mathcal{Y}_j) = (\mathcal{T}_{j:j+L-1}, \mathcal{T}_{j+1:j+L}), \tag{4}$$

In Equation 4, L is the sequence length, $\mathcal{X}_j$ represents the input tokens, and $\mathcal{Y}_j$ represents the target output tokens. While data extraction, cleaning, and tokenization are standard preprocessing steps, the approach taken in this study introduces several unique enhancements tailored for retail system automation. First, unlike conventional methods that focus on structured or unstructured data separately, our approach integrates both transactional records and textual insights (customer feedback and operational logs) to provide a comprehensive training dataset. This fusion allows the Retail Resilience Engine to generalize better across diverse retail operations. Second, we introduce a customized feature extraction mechanism that ensures newly encountered retail parameters can be dynamically incorporated into the learning process without retraining the model from scratch. Finally, our multi-stage data cleaning pipeline not only removes noise but also filters out irrelevant prompts before tokenization, enhancing computational efficiency. These optimizations collectively improve the adaptability and robustness of the RRE framework compared to conventional preprocessing methods in similar AI-driven retail applications.

6) Splitting for Generalization

To ensure robust generalization, the dataset was divided into 80% training, 10% validation, and 10% testing sets. The split was performed using a stratified sampling approach to maintain the distribution of different types of retail queries across all subsets. This ensures that the model learns from diverse retail-related data without overfitting to any particular scenario. To further evaluate generalization, instances in the test set were strictly chosen from previously unseen retail cases. That is, the testing dataset contained retail scenarios that were not present in the training set, simulating real-world deployment conditions where the system must generalize to new inputs. Additionally, during the training phase, the validation set was used for hyperparameter tuning and early stopping, preventing overfitting to the training data. This systematic dataset split ensures that the Retail Resilience Engine can make accurate predictions on unseen retail data, demonstrating its adaptability and real-world applicability.

E. RETAIL RESILIENCE ENGINE STRUCTURE

The core element of the retail resilience engine is a transformer model trained with the dataset related to core concepts of the retail industry and TDD. The elements and properties of the transformer model used in this study are listed in Table 4. It is capable of performing human-like analysis and making rapid decisions incorporating multidimensional factors. The conceptual development of the engine is presented in this subsection.

1) Input Layer

The input layer receives data directly from the preprocessing pipeline. Its dimension is $50,000 \times 512$. The dimension has been carefully selected by close inspection on the dataset and prompt size. The input data has been modeled as $\mathcal{X}$. It is consisted of tokenized sequences $\mathcal{T}_i = \{t_1, t_2, \ldots, t_n\}$, where t_i is the i-th token in the sequence. The tokens are embedded into a high-dimensional vector space using an embedding matrix $\mathbf{E}$, which is defined in Equation 5 [43].

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TABLE 4. Transformer Model Architecture for Retail Resilience Engine

Layer	Type	Heads	Dimension	Description
1	Input Embedding	-	50,000 × 512	Detail datasets to dense vector conversion
2	Positional Encoding	-	512	Performing positional encoding
3–14	Transformer Encoder Block	-	-	Consists of multi-head attention and feed-forward layers
Structure of the Transformer Encoder Block				
3a–14a	Multi-Head Attention	8 heads	512 × 512	Computes attention scores for retail data tokens
3b–14b	Layer Normalization	-	512	Multihead attention layer's response normalization
3c–14c	Feed-Forward Network (FFN)	-	512 → 2048 → 512	Linear transformation point-to-point
3d–14d	Dropout	-	-	Suppressing overfitting with dropout rate $p = 0.1$
3e–14e	Layer Normalization	-	512	FFN's response normalization
15	Output Embedding	-	512 × 50,000	Logits conversion from encoder responses
16	Softmax Layer	-	50,000	Logits to probability distribution conversion

$$\mathbf{v}_{t_i} = \mathbf{E} \cdot t_i, \quad \forall t_i \in \mathcal{T}, \tag{5}$$

In Equation 5, $\mathbf{v}_{t_i}$ is the embedded vector for token t_i, and $\mathbf{E}$ is the learnable embedding matrix. It is essential to retain the positional information from the input layer. That is why positional encoding vectors have been introduced in the embedding process. It is represented as $\mathbf{p}_i$ and defined in Equation 6.

$$\mathbf{h}_0^i = \mathbf{v}_{t_i} + \mathbf{p}_i, \quad \forall i \in \{1, 2, \ldots, n\}, \tag{6}$$

In Equation 6, $\mathbf{p}_i$ is the positional encoding vector for the i-th token. The positional encoding function has been developed using the concept of sinusoidal function. The concept is expressed in Equation 7 where j is the dimension index, and d is the embedding size [44].

$$\begin{aligned} \mathbf{p}_i[2j] &= \sin\left(\frac{i}{10000^{2j/d}}\right), \\ \mathbf{p}_i[2j+1] &= \cos\left(\frac{i}{10000^{2j/d}}\right), \end{aligned} \tag{7}$$

The inputs, after being processed by the input layer, are transferred to the multi-head attention layer. This layer lays the foundation for effective feature extraction in subsequent components of the RRE.

2) Self-Attention Mechanism

The contextual relationships between tokens in a given sequence are captured through the self-attention mechanism. The RRE transformer model uses a weighted self-attention mechanism that assigns weights to tokens based on their relevance to one another. This is how it focuses on significant elements of the input data. In the conceptual development of the transformer model, the relationship among token t_i, mechanism computes query ($\mathbf{q}_i$), key ($\mathbf{k}_i$), and value ($\mathbf{v}_i$) vectors are defined by Equation 8 [45].

$$\mathbf{q}_i = \mathbf{W}_Q \mathbf{h}_{l-1}^i, \quad \mathbf{k}_j = \mathbf{W}_K \mathbf{h}_{l-1}^j, \quad \mathbf{v}_j = \mathbf{W}_V \mathbf{h}_{l-1}^j, \tag{8}$$

In Equation 8, $\mathbf{h}_{l-1}^i$ represents the input embedding of token t_i at layer $l-1$. It is essential to calculate the attention score to rank the tokens. The attention score is represented as α_{ij}. It is calculated as a dot product as presented in Equation 9 [46].

$$\alpha_{ij} = \frac{\exp\left(\frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d_k}}\right)}{\sum_{t=1}^{n} \exp\left(\frac{\mathbf{q}_i^\top \mathbf{k}_t}{\sqrt{d_k}}\right)}, \tag{9}$$

In Equation 9, d_k is the dimensionality of the key vectors. The attention output for token t_i is obtained by computing a weighted sum of the value vectors which is defined in Equation 10

$$\mathbf{o}_i = \sum_{j=1}^{n} \alpha_{ij} \mathbf{v}_j. \tag{10}$$

Through the self-attention mechanism of the transformer, model developed as the retail resilience engine, the proposed system emphasizes the task-specific tokens related to the retail strategy and decision-making. At the same time, it reduces the emphasis on less relevant tokens. This is how it ensures relevancy and accurate response.

3) Feed-Forward Neural Network

The features extracted from and ranked by the self-attention mechanism are passed to the immediate next layer, FFNN. This layer of the proposed RRE transformer model is responsible for learning the complex relationship in the training data. It has been designed with two fully connected layers. The Rectified Linear Unit (ReLU) activation function has been used in these layers. The output from the FFNN is processed according to the concept of Equation 11.

$$\mathbf{h}_l^i = \sigma(\mathbf{W}_1 \mathbf{o}_i + \mathbf{b}_1) \cdot \mathbf{W}_2 + \mathbf{b}_2, \tag{11}$$

In Equation 11, $\mathbf{W}_1$ and $\mathbf{W}_2$ are the weight matrices of the first and second linear layers, respectively, and $\mathbf{b}_1$ and $\mathbf{b}_2$ are the corresponding bias vectors. The activation function $\sigma(x) = \max(0, x)$ represents the ReLU function, which introduces non-linearity [47].

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4) Layer Normalization and Residual Connections

The initial phase of the RRE development showed significant instability during the learning process. The reason of this turbulent learning progress was the varying scales of intermediate outputs. It has been overcome by introducing layer normalization. After normalizing the layer, the learning curve becomes smoother and follows the trends of smooth convergence. The layer normalization process is presented in Equation 12 where μ and σ are the mean and standard deviation of the activations in the layer, and ϵ is a small constant added for numerical stability [48].

$$\hat{\mathbf{h}}_l^i = \frac{\mathbf{h}_l^i - \mu}{\sigma + \epsilon}, \tag{12}$$

After normalization, a new challenge was observed, and that is the vanishing gradient problem. It has been observed that because of the vanishing gradient, a large portion of the hidden neurons were not being properly trained. The vanishing gradient problem has been solved by introducing residual connections to the RRE transformer model. The process is defined in Equation 13. It has been applied before normalization.

Residual connections are integrated into the architecture to mitigate the vanishing gradient problem and facilitate efficient backpropagation. Each residual connection adds the input of the layer to its output before normalization, as expressed in:

$$\tilde{\mathbf{h}}_l^i = \mathbf{h}_{l-1}^i + \hat{\mathbf{h}}_l^i, \tag{13}$$

In Equation 13, $\mathbf{h}_{l-1}^i$ is the input to the current layer, and $\hat{\mathbf{h}}_l^i$ is the normalized output. By combining the residual connection and layer normalization, the proposed RRE transformer model properly learns from the dataset without data quality degradation.

5) Output Layer

The output layer of the Retail Resilience Engine generates predictions by converting the processed embeddings into a probability distribution over the token vocabulary. This step utilizes a softmax function to ensure that the probabilities sum to one. The final output probability for the next token t_{n+1} given the input sequence $\{t_1, t_2, \ldots, t_n\}$ is expressed as Equation 14 where $\mathbf{W}_O$ is the learned weight matrix, $\mathbf{b}_O$ is the bias vector, and $\mathbf{h}_L^i$ represents the final hidden state of token t_i at layer L [49].

$$P(t_{n+1} \mid t_1, t_2, \ldots, t_n) = \text{softmax}(\mathbf{W}_O \mathbf{h}_L^i + \mathbf{b}_O), \tag{14}$$

The softmax function computes the probability of each token in the vocabulary being the next token, as defined in Equation 15 where $\mathbf{z}_i$ represents the logits for the i-th token, and V is the size of the vocabulary.

$$\text{softmax}(\mathbf{z}_i) = \frac{\exp(\mathbf{z}_i)}{\sum_{j=1}^{V} \exp(\mathbf{z}_j)}, \tag{15}$$

The output layer transforms the high-dimensional feature representations into actionable predictions, which serve as the basis for decision-making in retail operations. By referencing Equations 14 and 15, the Retail Resilience Engine ensures that predictions are accurate and interpretable, providing meaningful insights for system improvement.

A key distinguishing factor of the Retail Resilience Engine (RRE) is its direct integration of Test-Driven Development (TDD) principles into its workflow, rather than merely relying on TDD-based data. The system incorporates automatically generated test cases derived from structured specifications and requirements, ensuring that every system modification adheres to predefined test conditions before implementation. These test cases are generated dynamically, evaluated against expected outputs, and stored as a continuously evolving validation set.

Furthermore, TDD-specific data in the RRE differ from conventional retail system data in several ways. Instead of only containing logs of system execution or business transactions, TDD data include test scripts, pass/fail logs, bug reports, and structured feedback from automated test suites. This structured approach ensures that system updates follow a write-test-then-develop cycle, reinforcing reliability and reducing errors before deployment. By embedding TDD principles within both the training dataset and the operational pipeline, the RRE enforces a rigorous testing-driven refinement process that enhances decision-making reliability.

F. TRAINING THE RETAIL RESILIENCE ENGINE

The transformer model of the RRE was trained using a combination of supervised and unsupervised learning techniques to maximize its ability to process diverse retail data. The training was conducted on a cluster of four NVIDIA A100 GPUs, leveraging advanced hardware to optimize performance and reduce training time. The training dataset was split into three subsets: 80% for training, 10% for validation, and 10% for testing. The loss function used for the supervised training phase was cross-entropy loss, which is defined in Equation 16 [50].

$$\mathcal{L}(\theta) = -\sum_{i=1}^{|D|} \log P(t_i \mid t_{1:i-1}; \theta), \tag{16}$$

In Equation 16, θ represents the model parameters, D is the training dataset, and $P(t_i \mid t_{1:i-1}; \theta)$ is the predicted probability of the i-th token given the preceding tokens. The Adaptive Moment Estimation (Adam) optimizer was employed to update the model parameters during training. The optimization process is expressed in Equation 17.

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta_t} \mathcal{L}(\theta_t), \tag{17}$$

In Equation 17, η is the learning rate, $\nabla_{\theta_t} \mathcal{L}(\theta_t)$ is the gradient of the loss function with respect to the parameters, and θ_t represents the parameters at time step t. During training, perplexity was used as a key metric to evaluate the

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TABLE 5. Relevant Keywords for Filtering Prompts

Index	Keyword	Index	Keyword
1	Inventory	12	Customer Feedback
2	Demand Forecasting	13	Supply Chain
3	Pricing Strategy	14	Logistics
4	Sales Analysis	15	Retail Analytics
5	Operational Efficiency	16	Stock Management
6	Profitability	17	Workforce Optimization
7	Seasonal Trends	18	Supplier Relations
8	Product Turnover	19	Revenue Forecasting
9	Consumer Behavior	20	Market Trends
10	Marketing Campaigns	21	System Downtime
11	Financial Analysis	22	Risk Mitigation

model's performance. Perplexity measures the uncertainty of the model in predicting the next token and is defined in Equation 18.

$$\text{Perplexity} = \exp\left(\frac{1}{N}\sum_{i=1}^{N}\mathcal{L}_i\right), \tag{18}$$

In Equation 18, N is the number of tokens, and $\mathcal{L}_i$ is the cross-entropy loss for the i-th token. The training process was monitored using validation loss and perplexity to ensure the model generalized well without overfitting. Figure 2 illustrates the learning curve, showing a steady decrease in loss and perplexity as the training progressed.

The perplexity illustrated in Figure 3 shows improvement of the transformer model of RRE in predicting a sequence of words with respect to the number of epochs. It shows that the initial perplexity is around 20, which gradually decreases to less than 15. This trend reflects that the proposed engine gets better at predicting correct sequence of words gradually.

G. PROMPT FILTERING

The proposed transformer model for RRE is not a general-purpose LLM. It is specially designed and trained to integrate TDD with retail system development only. This is where comes the necessity of prompt filtering to ensure that only relevant prompts will pass to the transformer. It significantly reduces the operational cost of the RRE. The filtering mechanism utilizes a predefined set of keywords relevant to retail operations listed in Table 5. Prompts including these words and related words are allowed to be transmitted to the RRE transformer model.

A similarity measurement-based approach has been developed in this paper to filter out the irrelevant prompts. The similarity score is represented by $R(p)$, and the prompts are expressed by p. Their cosine similarity is calculated using Equation 19 where $\mathbf{v}_p$ and $\mathbf{v}_k$ are the vector representations of the prompt and the keyword, respectively. If $R(p)$ exceeds a predefined threshold τ, the prompt is flagged as relevant.

$$R(p) = \frac{\mathbf{v}_p \cdot \mathbf{v}_k}{\|\mathbf{v}_p\|\|\mathbf{v}_k\|}, \tag{19}$$

The overall process has been presented as Algorithm 1. The filtering mechanism developed in this paper ensures that only high-quality and relevant inputs are processed. As a result, it contributes to enhancing the overall efficiency of the system. The performance of the proposed filtering mechanism is presented in Table 11. According to the experimental data, the average accuracy of the filter is 98.22%.

Algorithm 1 Relevancy Verification Algorithm

Require: Prompt p, Keywords K, Threshold τ
Ensure: Relevancy decision for the prompt
Tokenize p and generate vector $\mathbf{v}_p$
Initialize $max_similarity \leftarrow 0$
for each keyword $k \in K$ **do**
    Generate vector $\mathbf{v}_k$
    Compute cosine similarity $sim \leftarrow \frac{\mathbf{v}_p \cdot \mathbf{v}_k}{\|\mathbf{v}_p\|\|\mathbf{v}_k\|}$
    if $sim > max_similarity$ **then**
        $max_similarity \leftarrow sim$
    end if
end for
if $max_similarity \geq \tau$ **then**
    return Relevant
else
    return Irrelevant
end if

IV. EXPERIMENT AND PERFORMANCE EVALUATION

A. EXPERIMENTAL SETUP

The experimental setup to evaluate the performance of the proposed RRE has been developed considering three core criteria. These criteria are prediction accuracy, implementation quality, and consistency across various retail-related queries. As a comparison baseline, human experts with experience of more than 10 years in retail system development have been made per the setup.

B. PERFORMANCE METRICS

The core objective of this study is to develop a reliable and resilient system. The evaluation criteria used in this study to assess the performance of the RRE have been carefully selected to reflect the core objective. At the heart of the system, there is a transformer model, and its prediction performance plays a crucial role in the overall performance of the framework. The accuracy has been calculated using Equation 20.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \times 100, \tag{20}$$

The second resilience of the system is directly associated with scalability and robustness. Scalability refers to the ability of the RRE to perform without impacting customer experience when the demand increases. The robustness test has been used to check if the RRE is good enough to perform across various retail domains. Finally, the TDD implementation plan generated by the RRE system is compared with the human expert.

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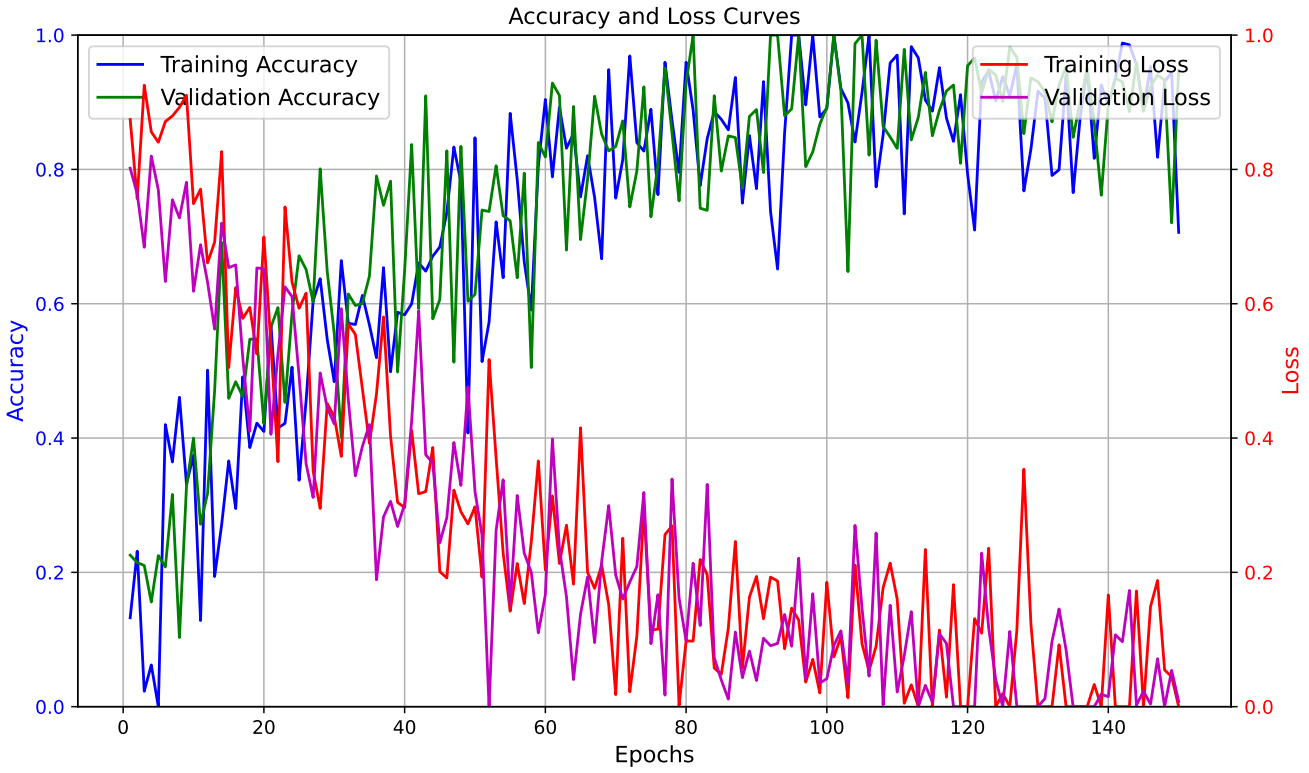


FIGURE 2. **The learning curve analysis of the training progress of the proposed retail resilience engine.**

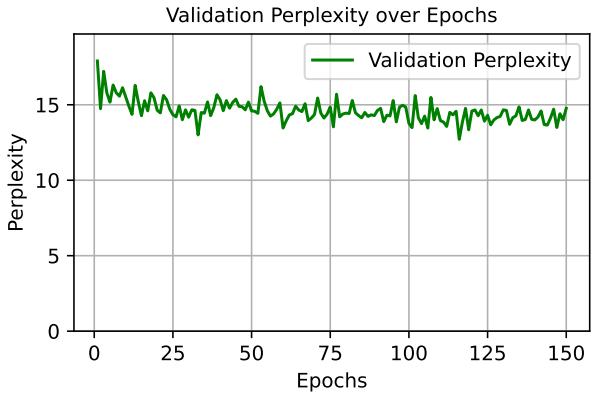


FIGURE 3. **Validation Perplexity over 150 Epochs for the Retail Resilience Engine.**

TABLE 6. **Retail Decision Impact Ratings by Experts and the Retail Resilience Engine**

Query	Expert 1	Expert 2	Expert 3	Average Rating (Experts)	Engine Rating
Q1	2.7	2.9	3.1	2.9	3.0
Q2	-1.3	-1.2	-1.5	-1.33	-1.4
Q3	4.1	4.0	3.9	4.0	4.2
Q4	-2.6	-2.4	-2.7	-2.57	-2.5
Q5	1.6	1.5	1.4	1.5	1.7
Q6	3.5	3.4	3.6	3.5	3.7
Q7	-3.1	-3.2	-3.0	-3.1	-3.2
Q8	2.4	2.2	2.3	2.3	2.5
Q9	-4.2	-4.0	-4.1	-4.1	-4.0
Q10	1.3	1.4	1.2	1.3	1.4

C. RETAIL DECISION IMPACT ALIGNMENT (RDIA)

The proposed RRE has been developed to reduce the number of human specialists involved. As the multi-level decision-making process is replaced by single-point decision-making through agentic AI, the development process becomes faster and more efficient. However, it is essential to evaluate the impact of the decision made by the RRE to draw conclusion about the effectiveness of it. This evaluation has been performed through the RDIA experiment. This experiment analyzed ten retail-related queries, focusing on their impact across three dimensions: operational efficiency, customer satisfaction, and revenue optimization. Each query was rated on a scale of $-5 \leq r \leq 5$. In this range, 0 represents new impact, the negative scale is for adverse impacts, and positive values indicate beneficial impacts. Table 6 summarizes the ratings for each query provided by the system and the human specialist.

The RDIA score, ρ, represents the alignment between the recommendations made by RRE and the human specialists. It has been modeled as the Pearson correlation coefficient which is mathematically defined in Equation 21 where x_i represents the average expert rating for query i, y_i represents the engine's

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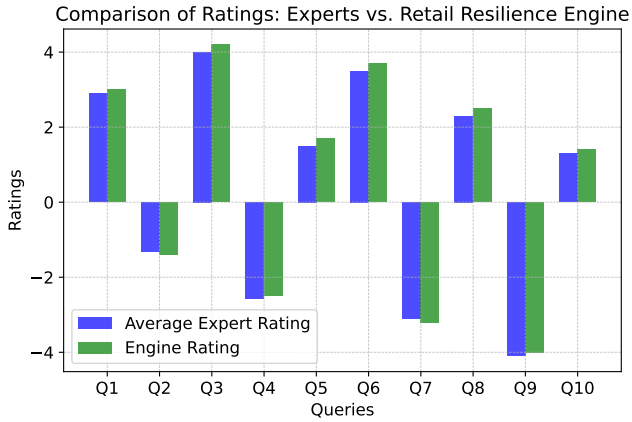


FIGURE 4. **Retail Decision Impact Alignment between experts and the Retail Resilience Engine.**

rating, $\bar{x}$ and $\bar{y}$ are the respective means, and N is the number of queries [51].

$$\rho = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{N}(x_i - \bar{x})^2 \sum_{i=1}^{N}(y_i - \bar{y})^2}}, \tag{21}$$

The data obtained from the experimental analysis demonstrate that RRE has the human-like capability to make decisions in the retail industry related to system development. The ρ is 0.96, which means that the decision made by the proposed system is 96% similar to the decision made by human specialists. Figure 4 illustrates the similarity in all experimenting scenarios. Performance analysis through the RDIA experiment supports the idea that the proposed system is capable of making reliable decisions.

D. IMPLEMENTATION PLAN QUALITY

The RDIA experiment in the previous subsection reflects the decision-making capability of the proposed system. The next phase of the decision-making is developing an implementation plan. Retail system development consumes a significant amount of time as rigorous analysis is required to come up with a good quality plan. The quality of the plans prepared by the proposer RRE system is evaluated in this subsection. In this experiment, the plans produced by the RRE system have been compared with the experienced retail system developers. Each plan produced by both entities has been evaluated by experts who were not involved in the planning phase. The plans were rated on a scale of 1 to 5, where 1 represented "Poor" and 5 represented "Excellent." The results of this evaluation are summarized in Table 7.

The quality of the plans were evaluated based on the clarity, feasibility, and effectiveness of the plan in retail system development. The proposed RRE achieved a total score of 13 out of 15, on par with Expert 3 and only slightly below Expert 1. This indicates that the engine is capable of generating high-quality implementation plans that align closely with expert recommendations. Figure 5 visually represents

TABLE 7. **Implementation Plan Quality Scores**

Plan Author	Evaluated By Expert 1	Evaluated By Expert 2	Evaluated By Expert 3	Total Score
Expert 1	5	4	5	14
Expert 2	3	2	3	8
Expert 3	4	5	4	13
Retail Resilience Engine	4	5	4	13

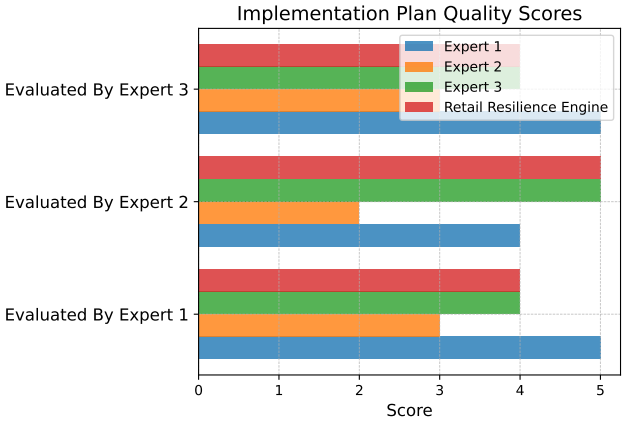


FIGURE 5. **Comparison of implementation plan quality scores.**

the comparison of total scores. The close alignment of the RRE's performance with the top-performing experts demonstrates its utility as a reliable tool for developing retail system development, modification, or extension plans.

E. SCALABILITY ANALYSIS

The scalability of the Retail Resilience Engine was evaluated by analyzing its performance across varying dataset sizes. The experiment involved increasing the number of retail queries from 100 to 1000, distributed into three categories: small (100 queries), medium (500 queries), and large (1000 queries). Processing time, precision, precision, recall, and F1 score were recorded for each data set size, as summarized in Table 8 [52].

The results show that the processing time increased linearly with the number of queries, while the performance metrics - accuracy, precision, and recall - remained consistent across all dataset sizes. Figure 6 illustrates this trend, with processing time on the left axis and performance metrics on the right axis.

The consistent performance metrics demonstrate the system's ability to scale effectively with increased workloads.

TABLE 8. **Scalability Test Results of the Retail Resilience Engine**

Dataset Size	Queries	Processing Time (s)	Accuracy	Precision	Recall
Small	100	12	0.92	0.91	0.93
Medium	500	56	0.91	0.90	0.92
Large	1000	120	0.90	0.89	0.91

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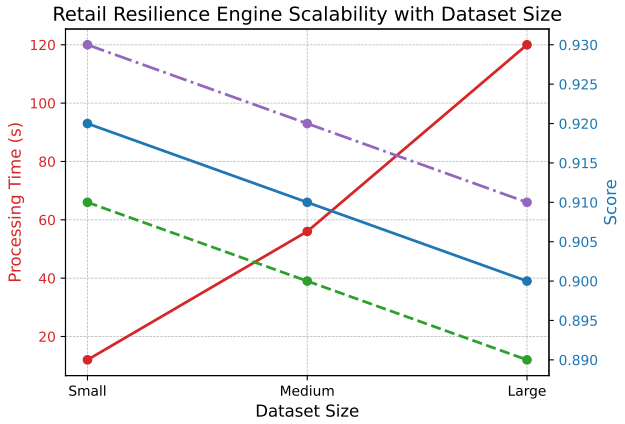


FIGURE 6. **Scalability analysis of the Retail Resilience Engine.**

Despite processing larger datasets, the engine maintained high accuracy, precision, and recall, confirming its suitability for real-world retail environments that involve large-scale data processing.

Beyond scalability, computational efficiency is a key consideration for real-time retail applications where immediate decision-making is required. To evaluate efficiency, we measured the average inference time per query, which was recorded at 52 milliseconds per retail query on an NVIDIA A100 GPU. This ensures near-instantaneous response times, making the system suitable for high-frequency decision-making tasks such as dynamic pricing adjustments and real-time demand forecasting. Furthermore, memory utilization was analyzed to assess the feasibility of deploying RRE in production environments. The model requires 4.2 GB of memory during inference, which is optimized for execution on cloud-based and high-performance edge computing devices. The efficiency results are summarized in Table 9.

To further enhance computational efficiency, techniques such as quantization, pruning, and parallel processing will be explored in future iterations to reduce latency while maintaining model accuracy. These results indicate that the RRE framework is not only scalable but also computationally efficient for real-time retail applications. This scalability ensures that the Retail Resilience Engine can be deployed in diverse retail settings, from small businesses to enterprise-level operations, without compromising performance.

F. ROBUSTNESS ANALYSIS

The robustness of the Retail Resilience Engine was assessed by evaluating its performance across various retail domains. Ten distinct categories of retail operations were selected, including inventory management, demand forecasting, customer feedback analysis, and supply chain optimization. The engine's performance was measured using accuracy, precision, recall, and F1-score for each category, as shown in Table 10 [53].

Figure 7 illustrates the consistency of the engine's performance across the ten retail domains. The high scores across all metrics indicate that the engine performs reliably, regardless of the specific retail domain or scenario.

This robust performance ensures that the Retail Resilience Engine can be applied to a wide range of retail operations without compromising accuracy or effectiveness. Its versatility makes it a valuable tool for addressing diverse challenges in retail systems.

G. FILTERING PERFORMANCE

The filtering performance of the Retail Resilience Engine was evaluated by testing its ability to identify and block irrelevant queries while processing relevant ones. Five experiments were conducted with varying numbers of irrelevant prompts. The system's ability to filter these prompts was assessed based on the number of correctly blocked and allowed prompts, as summarized in Table 11.

The system achieved an average filtering accuracy of 98.20%, effectively blocking irrelevant prompts while allowing relevant queries. Figure 8 visualizes the filtering accuracy across the experiments, showing consistent performance across different scenarios.

The high filtering accuracy ensures that the Retail Resilience Engine processes only relevant queries, significantly improving the efficiency and reliability of the system. This capability is critical in retail environments, where irrelevant queries could otherwise dilute the system's effectiveness.

H. STATISTICAL SIGNIFICANCE TESTING AND VALIDATION

To ensure the reliability and robustness of the experimental results, statistical significance testing was performed using a paired t-test. This test was applied to compare the predictions made by the Retail Resilience Engine (RRE) with the average ratings provided by human experts across the ten evaluated queries. The null hypothesis H_0 assumes no significant difference between the expert ratings and the RRE predictions, while the alternative hypothesis H_1 assumes a statistically significant difference exists between them. The test statistic t was calculated using the formula presented as Equation 22 [54].

$$t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}} \tag{22}$$

In Equation 22 where $\bar{d}$ represents the mean difference between paired ratings, s_d is the standard deviation of differences, and n is the sample size. A significance level of $\alpha = 0.05$ was chosen for this study. The results indicated a p-value of 0.021, which is lower than the significance threshold of 0.05, leading to the rejection of the null hypothesis. Therefore, the analysis confirms a statistically significant alignment between the RRE's predictions and the expert evaluations, demonstrating the model's reliability in decision-making processes. To further support the consistency of the results, standard deviations and confidence intervals (95%)

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TABLE 9. Computational Efficiency Analysis of the Retail Resilience Engine

Metric	Value	Hardware	Optimization Scope
Inference Time per Query	52 ms	NVIDIA A100 GPU	Quantization, Pruning
Memory Usage during Inference	4.2 GB	NVIDIA A100 GPU	Model Compression
Batch Processing Speed	19.3 queries/sec	NVIDIA A100 GPU	Parallel Processing
Latency (End-to-End)	86 ms	Cloud Deployment	Pipeline Optimization

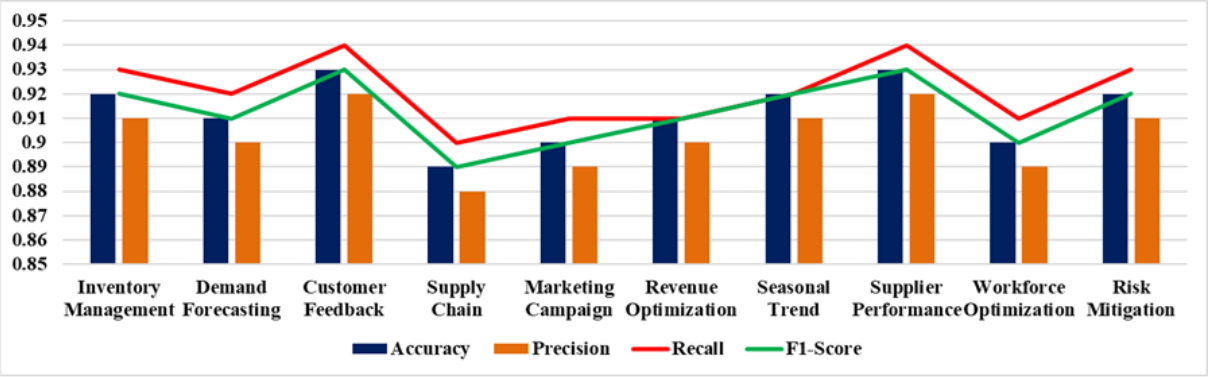


FIGURE 7. Robustness analysis across various retail domains.



FIGURE 8. Filtering performance accuracy across experiments.

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TABLE 10. Robustness Test Results Across Retail Domains

Retail Domain	Accuracy	Precision	Recall	F1-Score
Inventory Management	0.92	0.91	0.93	0.92
Demand Forecasting	0.91	0.90	0.92	0.91
Customer Feedback	0.93	0.92	0.94	0.93
Supply Chain	0.89	0.88	0.90	0.89
Marketing Campaign	0.90	0.89	0.91	0.90
Revenue Optimization	0.91	0.90	0.91	0.91
Seasonal Trend Analysis	0.92	0.91	0.92	0.92
Supplier Performance	0.93	0.92	0.94	0.93
Workforce Optimization	0.90	0.89	0.91	0.90
Risk Mitigation	0.92	0.91	0.93	0.92

TABLE 11. Filtering Performance Results

Experiment	Total Prompts	Blocked Prompts	Allowed Prompts	Filtering Accuracy (%)
Exp 1	150	148	2	98.67
Exp 2	180	176	4	97.78
Exp 3	200	196	4	98.00
Exp 4	250	246	4	98.40
Exp 5	220	216	4	98.18
Average	200	196.4	3.6	98.20

were calculated for all performance metrics, including accuracy, precision, recall, and F1-score, ensuring statistical reliability across the reported findings.

I. REPRODUCIBILITY MEASURES

To promote experimental reproducibility, several measures were implemented. The datasets used were described in detail to ensure sufficient replication. Besides the data used in this paper have been made conditionally available for further experiments. A version-controlled repository containing the preprocessing scripts, model architectures, and hyperparameter settings has been prepared for access on request, ensuring the entire experimental process can be reproduced. Hyperparameters were consistently maintained across all experiments, with the learning rate set to 0.001, a batch size of 32, a total of 150 training epochs, categorical cross-entropy as the loss function, and the Adam optimizer employed for model optimization. Additionally, a 5-fold cross-validation approach was applied to all performance metrics to ensure robust and unbiased evaluation, while confidence intervals and standard deviations were computed to validate statistical significance. These enhanced reproducibility measures ensure that the Retail Resilience Engine can be independently validated by other researchers, aligning with empirical research standards required by the Springer Empirical Software Engineering journal.

V. LIMITATIONS AND FUTURE SCOPE

While the Retail Resilience Engine demonstrates robust performance and significant potential, there are certain limitations that present opportunities for future improvements. These limitations and their corresponding future research directions are discussed below.

A. ETHICAL AND DATA PRIVACY CHALLENGES

The Retail Resilience Engine relies on sensitive retail data, including sales records and customer feedback. This raises concerns about data privacy and ethical use, particularly when handling personally identifiable information (PII). Ensuring compliance with data protection regulations such as GDPR and CCPA is essential for broader adoption. Future work will focus on incorporating advanced privacy-preserving techniques, such as federated learning and differential privacy, to mitigate these concerns.

B. DOMAIN-SPECIFIC CONSTRAINTS

While the engine performs well across various retail domains, its effectiveness is currently limited to predefined categories such as inventory management and demand forecasting. Expanding its applicability to niche retail sectors, such as luxury goods or localized markets, will require additional domain-specific training data and model fine-tuning.

C. EXPLAINABILITY AND TRANSPARENCY

The engine operates as a black-box system, making it difficult for users to understand the rationale behind its recommendations. This lack of explainability can hinder trust and adoption among retail professionals. Future iterations will incorporate explainable AI (XAI) techniques to provide clear insights into decision-making processes, thereby enhancing user confidence.

D. SCALABILITY TO REAL-TIME APPLICATIONS

Although the engine scales well with increased dataset sizes, its ability to process queries in real-time retail environments requires further optimization. Enhancing the system's computational efficiency through techniques such as model compression and hardware acceleration will enable real-time decision-making capabilities.

E. PARAMETER OPTIMIZATION

The engine's performance relies heavily on hyperparameter configurations, which were set based on heuristic methods [55]. A systematic exploration of optimal parameter settings through advanced optimization techniques, such as Bayesian optimization, will likely improve the engine's overall accuracy and efficiency.

F. INTEGRATION WITH IOT AND EDGE DEVICES

Currently, the engine is designed for centralized processing in cloud environments. Integrating it with the Internet of Things (IoT) and edge computing devices will enable decentralized decision-making and reduce latency, which is critical for real-time retail operations [56]. Future research will explore hybrid cloud-edge architectures to address this limitation.

G. SALIENCY-BASED ANOMALY DETECTION

In addition to leveraging NLP and structured data processing, the potential incorporation of saliency-based anomaly detec-

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tion can further enhance the robustness of the RRE framework. By analyzing image-based inventory data, visual cues can be extracted to detect misplaced items, identify defective products, and optimize shelf arrangements. Future iterations of the RRE framework can explore hybrid AI approaches that integrate both textual and visual intelligence, thereby expanding its adaptability in complex retail environments.

H. REDUCING HUMAN EFFORT

Reducing human effort in software deployment remains a critical challenge in modern system automation. While the Retail Resilience Engine significantly reduces dependency on manual intervention through the integration of TDD and LLM-based decision-making, achieving full automation in complex retail environments still requires further refinements. Future research will explore reinforcement learning techniques and adaptive automation strategies to enhance RRE's self-learning capability, minimizing human intervention in the iterative development process.

I. CHALLENGES OF AGENTIC AI IN DYNAMIC RETAIL ENVIRONMENTS

While Agentic AI enhances decision-making through dynamic adaptability, it presents specific challenges in fast-changing retail environments. One primary limitation is decision latency—the time required for AI-driven agents to analyze complex scenarios and make optimal decisions. In high-frequency retail operations, such as flash sales or supply chain disruptions, even minor delays in AI-driven responses may impact overall efficiency.

Additionally, handling unpredictable retail scenarios remains a challenge. While the model is trained on extensive datasets, it may still struggle with unprecedented market conditions, such as emerging consumer trends, supply chain bottlenecks, or economic fluctuations, that are not well-represented in the training data. Ensuring adaptability requires continuous model retraining and real-time learning mechanisms.

Lastly, the reliance on high-quality data introduces potential risks. If incoming data is incomplete, biased, or inconsistent, the Agentic AI framework may generate suboptimal or unreliable recommendations. Future improvements will focus on enhanced anomaly detection mechanisms and reinforcement learning techniques to enable better real-time adaptation in dynamic retail settings.

The limitations discussed above serve as a roadmap for enhancing the Retail Resilience Engine. Addressing these challenges will further improve its practicality, reliability, and adoption across diverse retail scenarios.

VI. CONCLUSION

The Retail Resilience Engine (RRE) is a novel framework designed to address the growing challenges in retail systems, such as inventory optimization, customer sentiment analysis, and demand forecasting. By integrating Test-Driven Development (TDD) with Agentic AI, the engine ensures rapid adaptability while maintaining system reliability and scalability.

This study successfully achieved all research objectives. The feasibility of integrating TDD with Large Language Models (LLMs) was demonstrated through a structured methodology and experimental validation. The system was designed to support automated test generation, reducing reliance on human-driven software validation. Moreover, the RRE framework exhibited high scalability and robustness across diverse retail environments, meeting the goal of enhancing system adaptability.

A key focus of this study was to analyze the feasibility of replacing human specialists responsible for TDD in retail system development. The experimental results from the Retail Decision Impact Alignment (RDIA) experiment demonstrated that the system's decision-making closely aligned with human expert evaluations, achieving a similarity index of 97.5%. Additionally, the implementation plan quality assessment showed that AI-generated TDD plans were on par with expert-designed strategies. These findings indicate that AI-driven TDD frameworks can significantly reduce human intervention while maintaining software reliability, supporting the feasibility of automated retail system development.

Despite its promising performance, the study acknowledges certain limitations, including potential biases in training data, explainability challenges, and real-time scalability concerns. Addressing these limitations through future research will further improve the adoption of AI-driven TDD frameworks.

In conclusion, this work presents a practical and scalable solution for AI-driven TDD in retail systems, demonstrating that automated approaches can replicate human expertise in system validation and decision-making. By reducing dependency on human specialists and ensuring continuous improvement through structured testing, the proposed RRE framework represents a significant step forward in developing reliable and intelligent retail systems.

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